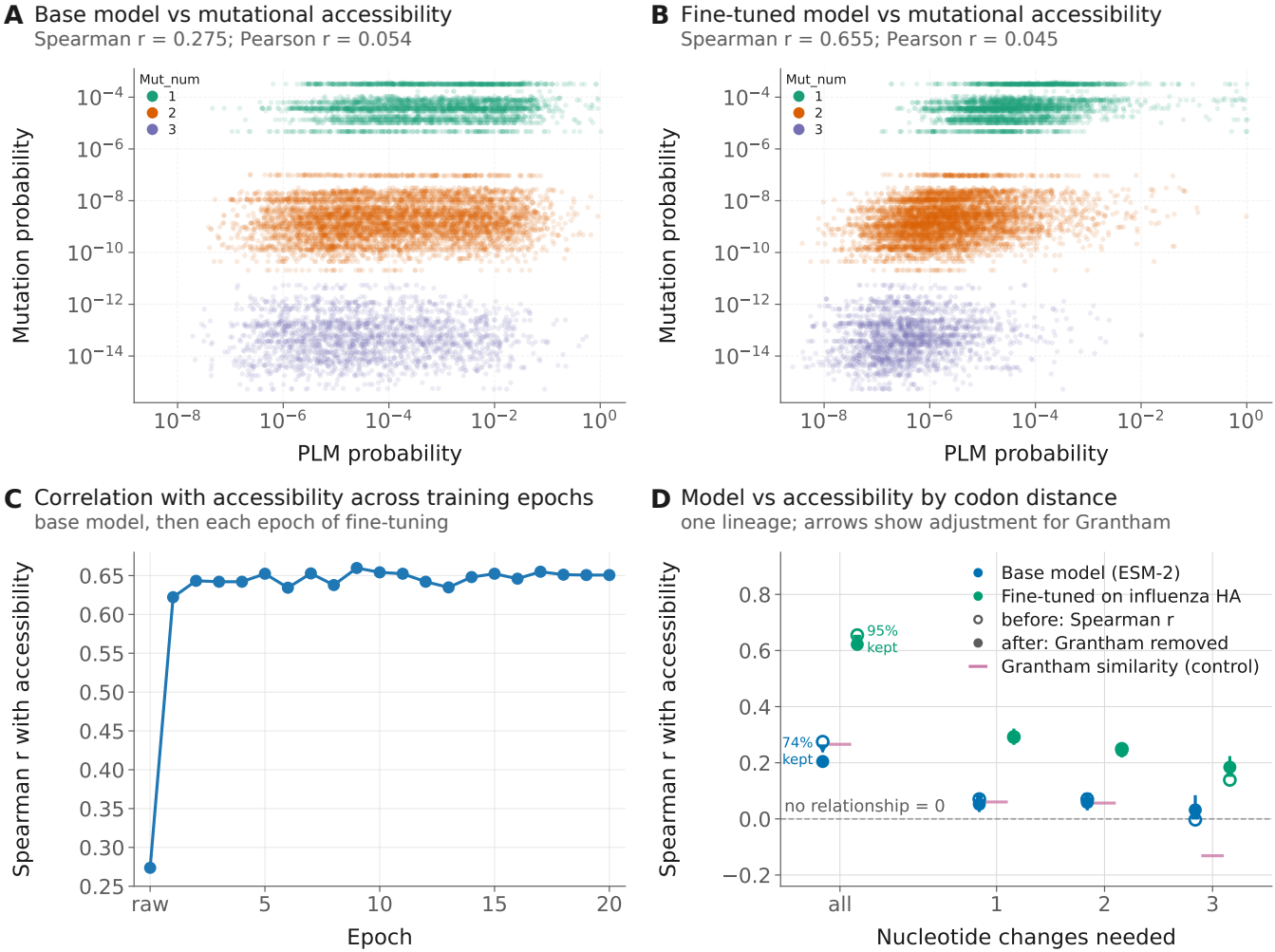

A Base model vs mutational accessibility
Spearman r = 0.275; Pearson r = 0.054
B Fine-tuned model vs mutational accessibility
Spearman r = 0.655; Pearson r = 0.045
Mut_num
1
2
3
Mutation probability
PLM probability
C Correlation with accessibility across training epochs
base model, then each epoch of fine-tuning
Spearman r with accessibility
Epoch
raw
D Model vs accessibility by codon distance
one lineage; arrows show adjustment for Grantham
Base model (ESM-2)
Fine-tuned on influenza HA
before: Spearman r
after: Grantham removed
Grantham similarity (control)
95% kept
74% kept
no relationship = 0
Nucleotide changes needed
all